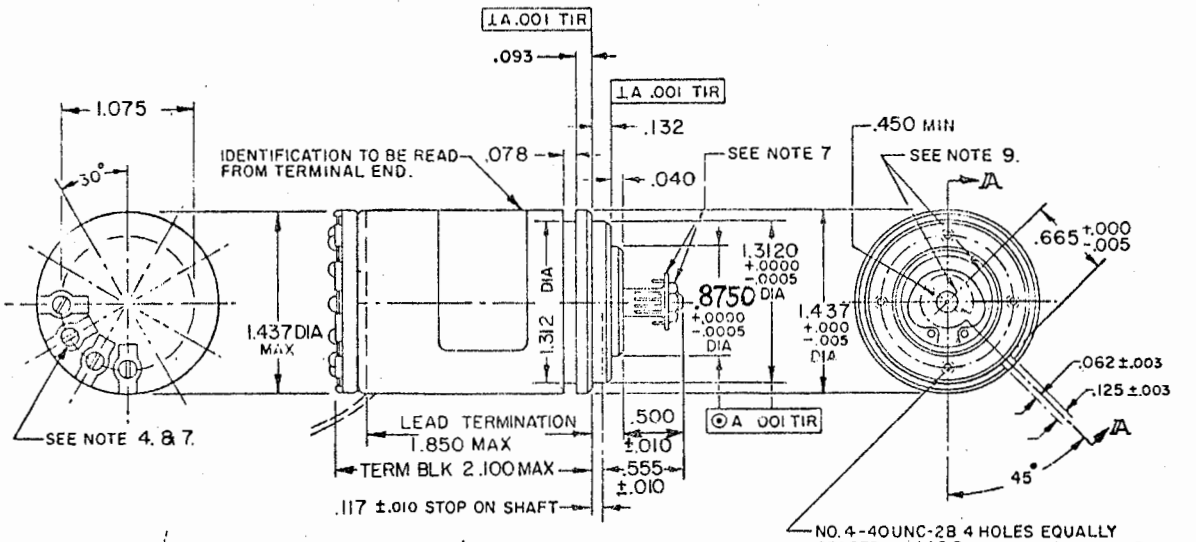
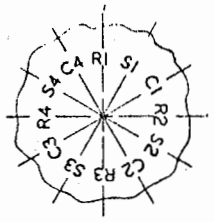
MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL

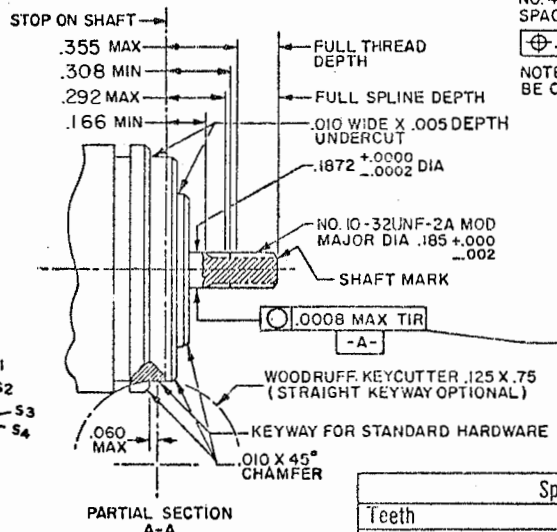
TYPE 15R28W4a

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of MIL-R-23417.

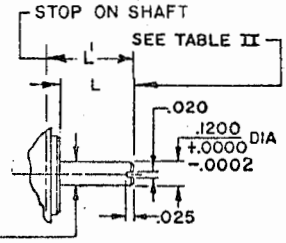




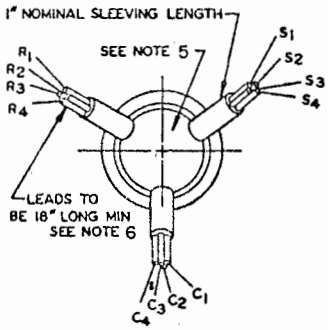
END VIEW- ENLARGED



PARTIAL SECTION A-A



PLAIN AND SLOTTED SHAFT



LEAD TERMINATION

Spline data	
Teeth	21
Pitch	120
Pitch diameter	.175 +.000, -.002
Outside diameter	.1872 +.0000, -.0005
Pressure angle	20°
Root diameter maximum	.155
Tooth form	Full depth involute
Note: The end function of this spline is for positive rotational fastenings. It is not to be used as a gear. This spline shall be inspected by (spline) go composite ring gage MS 90406.	

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are .XXX ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be shown and identified by molded lettering on the end cap or permanently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Mounting screw positions optional. Terminal block designed for MS 17182 terminal lug.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. Leads shall conform to MIL-W-16878/4, type E-28.
7. The following list of hardware shall be packaged in a ventilated envelope made of material having a PH value of 6-8 and compatable with the weight of the contents. The packaged hardware shall be placed in the unit resolver as follows:

Machine screw - - - - - -	(MS 35275-213)	12 Required (*) (**)
Lockwasher - - - - - - -	(MS 35338-135)	12 Required (*) (**)
Drivewasher - - - - - - -	(MS 17186-6)	1 Required (*)
Nut - - - - - - - - - - - -	(MS 17187-2)	1 Required (*)

(*) For units with spline shaft and terminal board termination.
(**) For units with plain smooth shaft and terminal board termination.

8. For information only: Use mounting clamp MS 17183 or equal.
9. Mark on housing is to match shaft mark within 10° when rotor is set at approximately resolver zero, and will bear no relation to the center lines shown on the drawing.
10. A voltage value of .5 to 26 volts shall be entered in the 'volts' designation space on the identification plate.
11. The weight shall be attached to the end of each wire, pulling straight away from the resolver. The lead shall then be bent 90° at the point of exit from the housing and while in this attitude the resolver rotated 360° counterclockwise. Any indication of the lead pulling loose, wire strands breaking, or permanent deformation of the insulation shall be cause for rejection under qualification tests.
12. With the resolver energized under standard test conditions the resonant frequency will be recorded during qualification as that point where the output voltage has a maximum value and phase shift - 90°. Low corner frequency (referenced as design information) is the point where output voltage is down 3 db and phase shift 45°.
13. T.R. variation and phase shift variation (rotor/compensator) will be tested as follows:

    Qualification test - Variation shall remain within specification limits when subjected to individual or simultaneous variations of voltage (.8 to 10 volts), frequency (380 to 420 Hz), and temperature (-55° C to +125° C). Test by applying input voltages of .8, 5, 10 volts with frequency and temperature extremes simultaneously applied.

    Acceptance test - Test by applying input voltages of .8, 5, 10 volts at rated frequency and room ambient temperatures.

14. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the pre-production tests as specified in MIL-R-23417.

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TABLE I

RESOLVER, ELECTRICAL: TYPE 15R28W4a

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

Requirement	Value	Unit	Tolerance
1. Excitation winding	Stator	---	---
2. Rated frequency	400	Hz	±1%
3. Input voltage:			
Maximum	26	Volts rms	±1%
Test	10	Volts rms	±1%
Minimum	0.5	Volts rms	±1%
4. Primary current @ test voltage	4	Milliamps	Nominal
5. Primary power	.006	Watts	Nominal
6. DC resistance:			
Rotor	200	Ohms	±15%
Stator	295	Ohms	±15%
Compensator	295	Ohms	±15%
7. Impedance:			
Zro	470 +j2990	Ohms	R±15%, +jx±10%
Zso	540 +j2800	Ohms	R±15%, +jx±10%
Zrs	520 +j250 (Reference)	Ohms	---
Zco	540 +j2800	Ohms	R±15%, +jx±10%
Zcs	530 +j21 (Reference)	Ohms	---
8. Resonant frequency (see note 12):			
Rotor	60	kHz	Minimum
Compensator	100	kHz	Minimum
9. Low corner frequency	42	Hz	Nominal
10. Transformation ratio:			
Output/input	.980	---	±1%
Rotor/compensator	.985	---	±.1%
11. T.R. variation (Rotor/compensator) due to voltage with test voltage as reference	±.05	Percent	Maximum
12. Equality of transformation ratio:			
Output/input	±.08	Percent	Maximum
Rotor/compensator	±.08	Percent	Maximum
13. Phase shift:			
Output/input	6	Degrees	±1.0
Rotor/compensator	0	Degrees	±.20
14. Phase shift variation (Rotor/compensator) due to voltage with test voltage as reference	±4	Minutes	Maximum
15. Interaxis error	3	Minutes	Maximum
16. Accuracy			
Functional error	±.1	Percent	Maximum
17. Null voltage @ test voltage:			
Fundamental	5	Mv rms	Maximum
Total	10	Mv rms	Maximum
18. Compensation			
Equality of compensating winding	20	Millivolts	Maximum
19. Shift of resolver zero:			
With voltage	±1	Minutes	Maximum
With frequency	±1	Minutes	Maximum
20. Harmonic distortion	0.1	Percent	Maximum
21. Temperature rise	15	° C	Maximum

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TABLE I - Continued

Requirement	Value	Unit	Tolerance
22. Friction torque:	.2	Oz-In	Maximum
@ -55° C	.4	Oz-In	Maximum
23. Radial play	.0005 (For reversal of 1/2 lb load)	Inches	Maximum
24. End play	.0010 (For reversal of 1 lb load)	Inches	±.0005
25. Wire lead stress (see note 11)	2	Pound	Minimum

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from table II.

Example:

M23417/12-01

Basic document number

Dash number from table II

TABLE II

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				L (±.010)	L' - Dim to stop on shaft (±.010)
-01	.187	Spline	Terminal board	.500 (1/2)	.555
-02	.120	Plain	Wire leads	.250 (1/4)	.305
-03	.120	Plain	Wire leads	.437 (7/16)	.492
-04	.120	Plain	Wire leads	.500 (1/2)	.555
-05	.120	Plain	Terminal board	.250 (1/4)	.305
-06	.120	Plain	Terminal board	.437 (7/16)	.492
-07	.120	Plain	Terminal board	.500 (1/2)	.555

Custodians:
Army - MU
Navy - AS
Air Force - 11

Review activities:
Army - MU, EL
Navy - AS, SH
Air Force - 11, 17, 85

User activities:
Army - MI, ME, AV
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - MU

(Project 5990-0117)

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